

KON 313

Feedback Control Systems

Linearization

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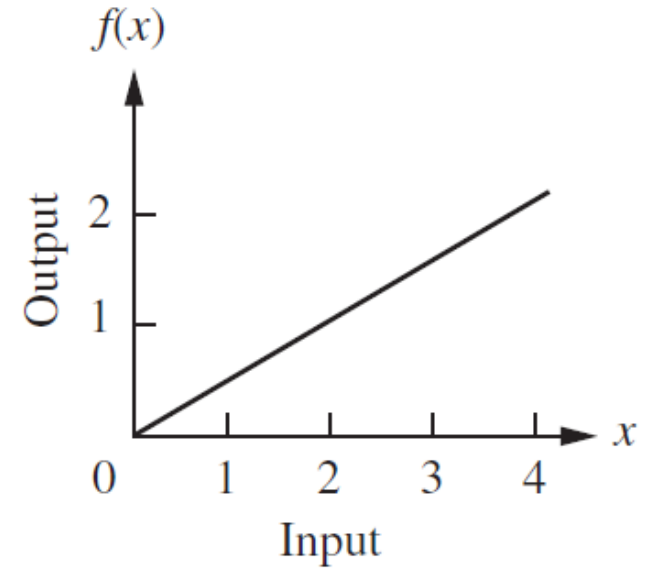
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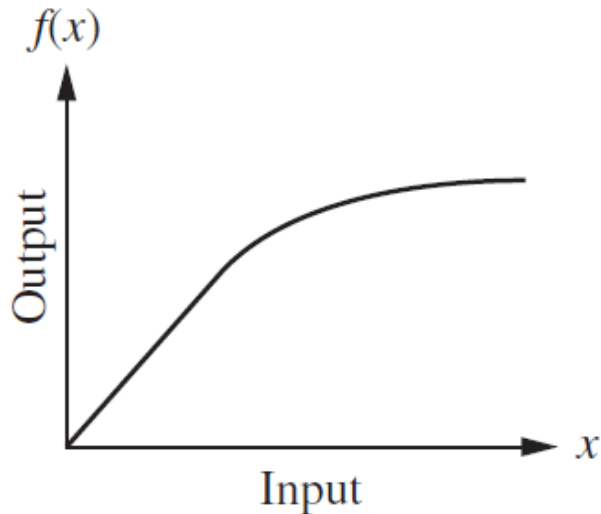
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Nonlinearity

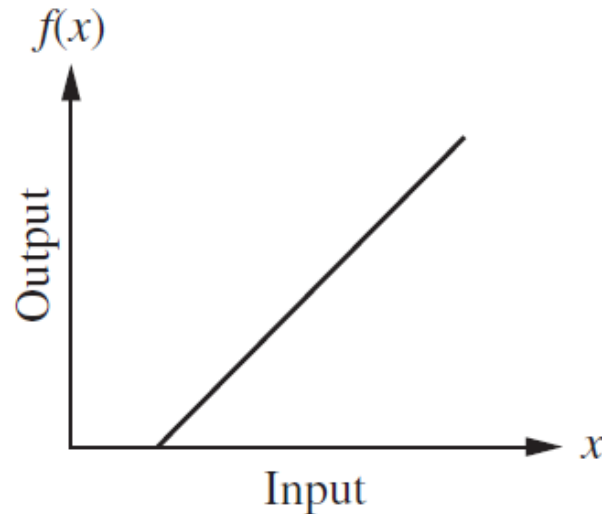
- A linear system possesses two properties: *superposition* and *homogeneity*.



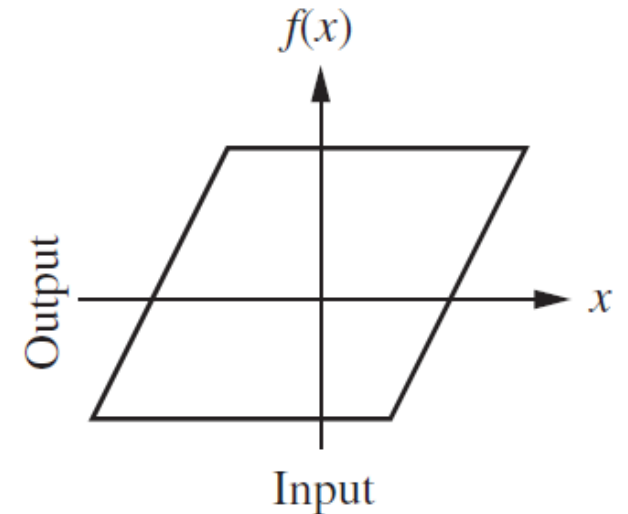
Amplifier saturation



Motor dead zone



Backlash in gears



Linearization

- The electrical and mechanical systems covered thus far were assumed to be linear.
- By approximating a nonlinear system as a linear system, we can use the modelling techniques covered so far.
- If any nonlinear components are present, we must linearize the system before we can find the transfer function.

Linearization

- The first step is to recognize the nonlinear component and write the **nonlinear differential equation**.
- We linearize a nonlinear differential equation around the operating points (**equilibrium** points, or steady states of the system). This point is selected as the second step in the linearization process. For example, when a pendulum is at rest, it is at equilibrium.
- Next we **linearize** the nonlinear differential equation.
- Then we take the **Laplace transform** of the linearized differential equation, assuming zero initial conditions.
- Finally, we separate input and output variables and form the **transfer function**.

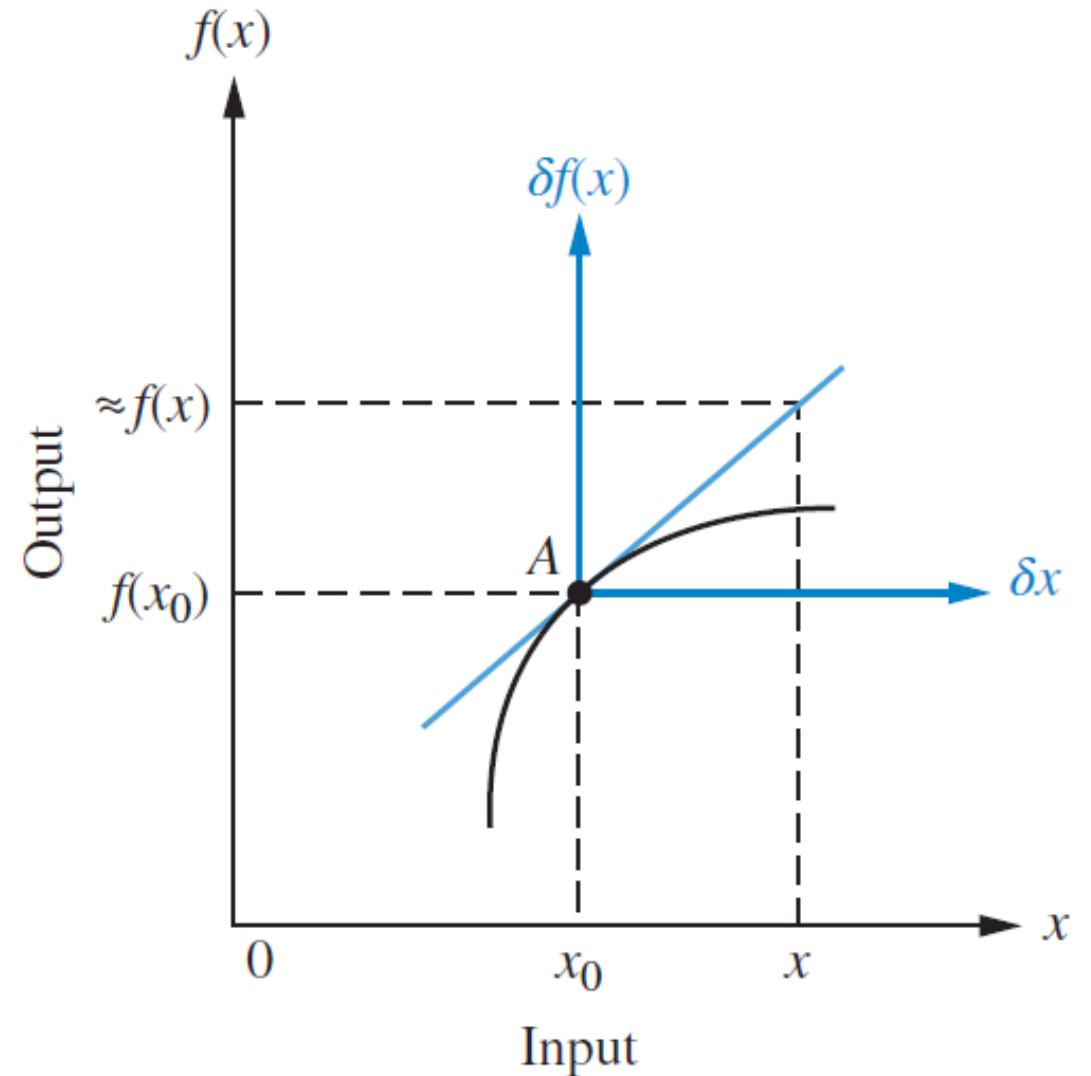
Linearization

- Given a nonlinear system operating at point A , $[x_0, f(x_0)]$, then small excursions of the input about point A , δx , can be related to small changes in the output about the point, $\delta f(x)$, by way of the **slope** of the curve at the point A , m_a ,

$$[f(x) - f(x_0)] \approx m_a(x - x_0)$$

$$\delta f(x) \approx m_a \delta x$$

$$f(x) \approx f(x_0) + m_a(x - x_0) \approx f(x_0) + m_a \delta x$$



Linearization

Taylor Series expansion

$$f(x) = f(x_0) + \left. \frac{df}{dx} \right|_{x=x_0} \frac{(x - x_0)}{1!} + \left. \frac{d^2f}{dx^2} \right|_{x=x_0} \frac{(x - x_0)^2}{2!} + \dots$$

- For small excursions of x from x_0 , we can neglect higher-order terms.

$$f(x) - f(x_0) \approx \left. \frac{df}{dx} \right|_{x=x_0} (x - x_0)$$

$$\delta f(x) \approx m \big|_{x=x_0} \delta x$$

which is a linear relationship between $\delta f(x)$ and δx for small excursions away from x_0 .

Linearization

Ex. Linearizing a Differential Equation

Linearize the following equation for small excursions about $x = \pi/4$.

$$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + \cos x = 0$$

Soln. The presence of the term $\cos x$ makes this equation nonlinear.

Since we want to linearize about $x = \pi/4$, we let $x = \delta x + \pi/4$, where δx is the small excursion about $\pi/4$. Substituting

$$\frac{d^2\left(\delta x + \frac{\pi}{4}\right)}{dt^2} + 2\frac{d\left(\delta x + \frac{\pi}{4}\right)}{dt} + \cos\left(\delta x + \frac{\pi}{4}\right) = 0$$

Linearization

Ex. Linearizing a Differential Equation (2)

Soln. (Cont.)

$$\frac{d^2 \left(\delta x + \frac{\pi}{4} \right)}{dt^2} + 2 \frac{d \left(\delta x + \frac{\pi}{4} \right)}{dt} + \cos \left(\delta x + \frac{\pi}{4} \right) = 0$$

$$\frac{d^2 \left(\delta x + \frac{\pi}{4} \right)}{dt^2} = \frac{d^2 \delta x}{dt^2} \quad \text{and} \quad \frac{d \left(\delta x + \frac{\pi}{4} \right)}{dt} = \frac{d \delta x}{dt}$$

The term $\cos(\delta x + \pi/4)$ can be linearized with the truncated Taylor series.

$$\cos \left(\delta x + \frac{\pi}{4} \right) - \cos \left(\frac{\pi}{4} \right) = \left. \frac{d \cos x}{dx} \right|_{x=\frac{\pi}{4}} \delta x = -\sin \left(\frac{\pi}{4} \right) \delta x$$

$$\cos \left(\delta x + \frac{\pi}{4} \right) = \cos \left(\frac{\pi}{4} \right) - \sin \left(\frac{\pi}{4} \right) \delta x = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} \delta x$$

Linearization

Ex. Linearizing a Differential Equation (3)

Soln. (Cont.)

$$\frac{d^2 \left(\delta x + \frac{\pi}{4} \right)}{dt^2} + 2 \frac{d \left(\delta x + \frac{\pi}{4} \right)}{dt} + \cos \left(\delta x + \frac{\pi}{4} \right) = 0$$

the following linearized differential equation is achieved:

$$\frac{d^2 \delta x}{dt^2} + 2 \frac{d \delta x}{dt} - \frac{\sqrt{2}}{2} \delta x = -\frac{\sqrt{2}}{2}$$

This equation can now be solved for δx , from which we can obtain $x = \delta x + \pi/4$.

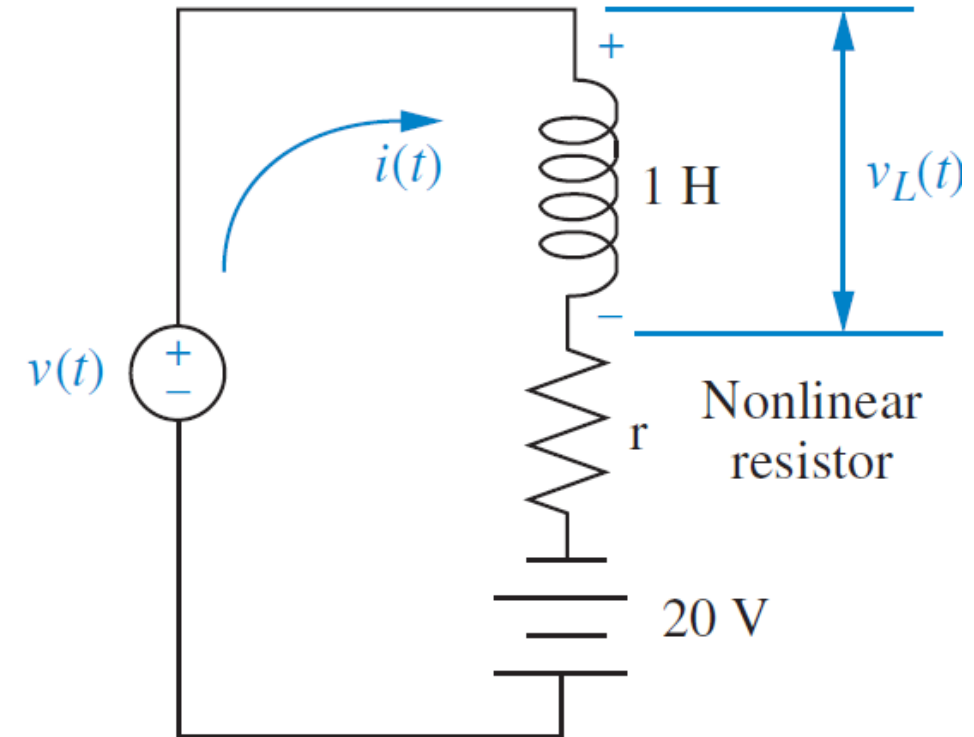
Linearization

Ex. Transfer Function—Nonlinear Electrical Network

Find the transfer function, $V_L(s)/V(s)$, for this electrical network, which contains a nonlinear resistor whose voltage-current relationship is defined by $i_r = 2e^{0.1v_r}$, where i_r and v_r are the resistor current and voltage, respectively. Also, $v(t)$ is a small-signal source.

Sln. Step1. Obtain the nonlinear differential equation. KVL:

$$v_r = 10 \ln \frac{1}{2} i_r, \text{ where } i_r = i, \quad \Rightarrow \quad L \frac{di}{dt} + 10 \ln \frac{1}{2} i - 20 = v(t)$$



Linearization

Ex. Transfer Function—Nonlinear Electrical Network (2)

Soln. (Cont.) Step2. Evaluate the equilibrium solution.

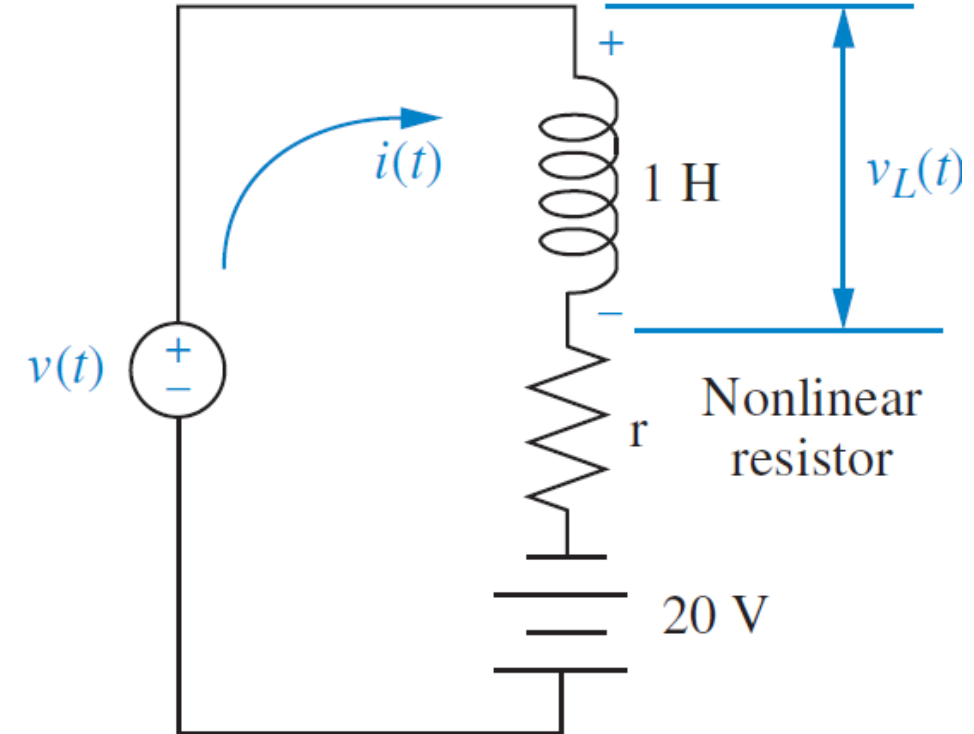
Set the small-signal source, $v(t) = 0$, and evaluate the steady-state current, $\frac{di}{dt} = 0$.

$$10 \ln \frac{1}{2} i = 20 \Rightarrow i_0 = 14.78 \text{ amps}$$

$$i = i_0 + \delta i$$

Thus

$$L \frac{d(i_0 + \delta i)}{dt} + 10 \ln \frac{1}{2} (i_0 + \delta i) - 20 = v(t) \quad \leftarrow \quad L \frac{di}{dt} + 10 \ln \frac{1}{2} i - 20 = v(t)$$



Linearization

Ex. Transfer Function—Nonlinear Electrical Network (3)

Soln. (Cont.)

$$L \frac{d(i_0 + \delta i)}{dt} + 10 \ln \frac{1}{2} (i_0 + \delta i) - 20 = v(t)$$

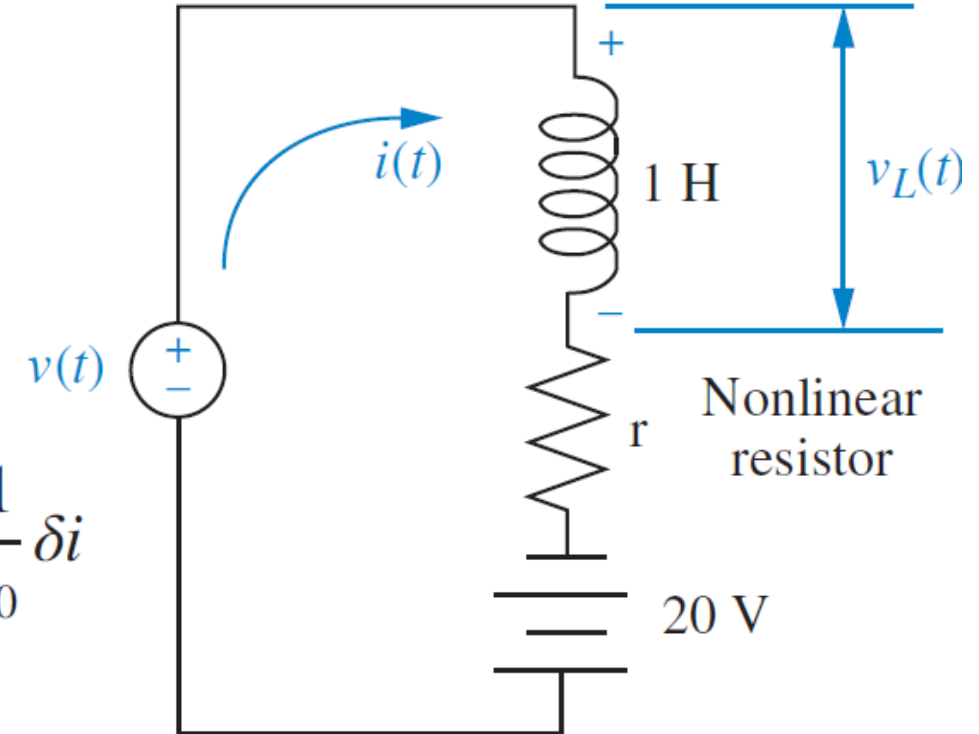
Step3. Linearize $\ln \frac{1}{2} (i_0 + \delta i)$

$$\ln \frac{1}{2} (i_0 + \delta i) - \ln \frac{1}{2} i_0 = \left. \frac{d \left(\ln \frac{1}{2} i \right)}{di} \right|_{i=i_0} \delta i = \left. \frac{1}{i} \right|_{i=i_0} \delta i = \frac{1}{i_0} \delta i$$

$$\ln \frac{1}{2} (i_0 + \delta i) = \ln \frac{i_0}{2} + \frac{1}{i_0} \delta i$$

Thus,

$$L \frac{d\delta i}{dt} + 10 \left(\ln \frac{i_0}{2} + \frac{1}{i_0} \delta i \right) - 20 = v(t) \quad \xRightarrow{i_0=14.78, L=1} \quad \frac{d\delta i}{dt} + 0.677 \delta i = v(t)$$



Linearization

Ex. Transfer Function—Nonlinear Electrical Network (4)

Soln. (Cont.) Step4. Take Laplace Transform.

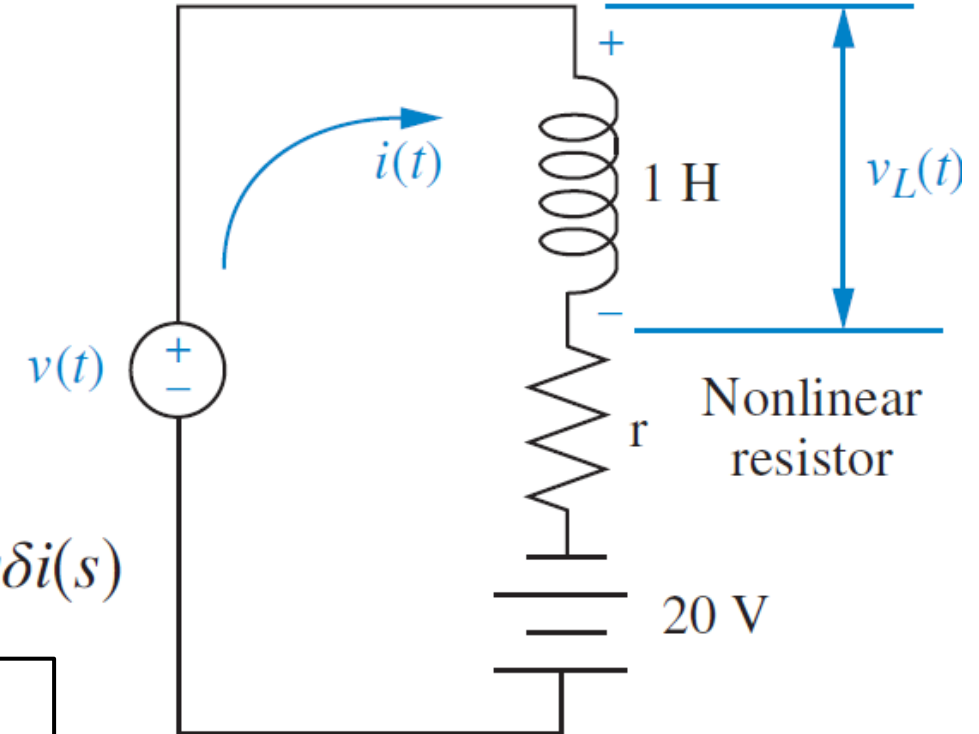
$$\frac{d\delta i}{dt} + 0.677\delta i = v(t) \quad \xRightarrow{\mathcal{L}} \quad \delta i(s) = \frac{V(s)}{s + 0.677}$$

Finally, find the transfer function.

$$v_L(t) = L \frac{d}{dt}(i_0 + \delta i) = L \frac{d\delta i}{dt} \quad \xRightarrow{\mathcal{L}} \quad V_L(s) = Ls\delta i(s) = s\delta i(s)$$

$$V_L(s) = s \frac{V(s)}{s + 0.677} \quad \Rightarrow \quad \boxed{\frac{V_L(s)}{V(s)} = \frac{s}{s + 0.677}}$$

for small excursions about $i = 14.78$ or, equivalently, about $v(t) = 0$.



Linearization

Ex. Transfer Function—Nonlinear System, simple pendulum

Mg is the weight, T is an applied torque in the θ direction, and L is the length of the pendulum. Assume the mass is evenly distributed, with the centre of mass at $L/2$. The pendulum's equilibrium point is the vertical position with zero angular velocity. Find the transfer function $\frac{\Theta(s)}{T(s)}$

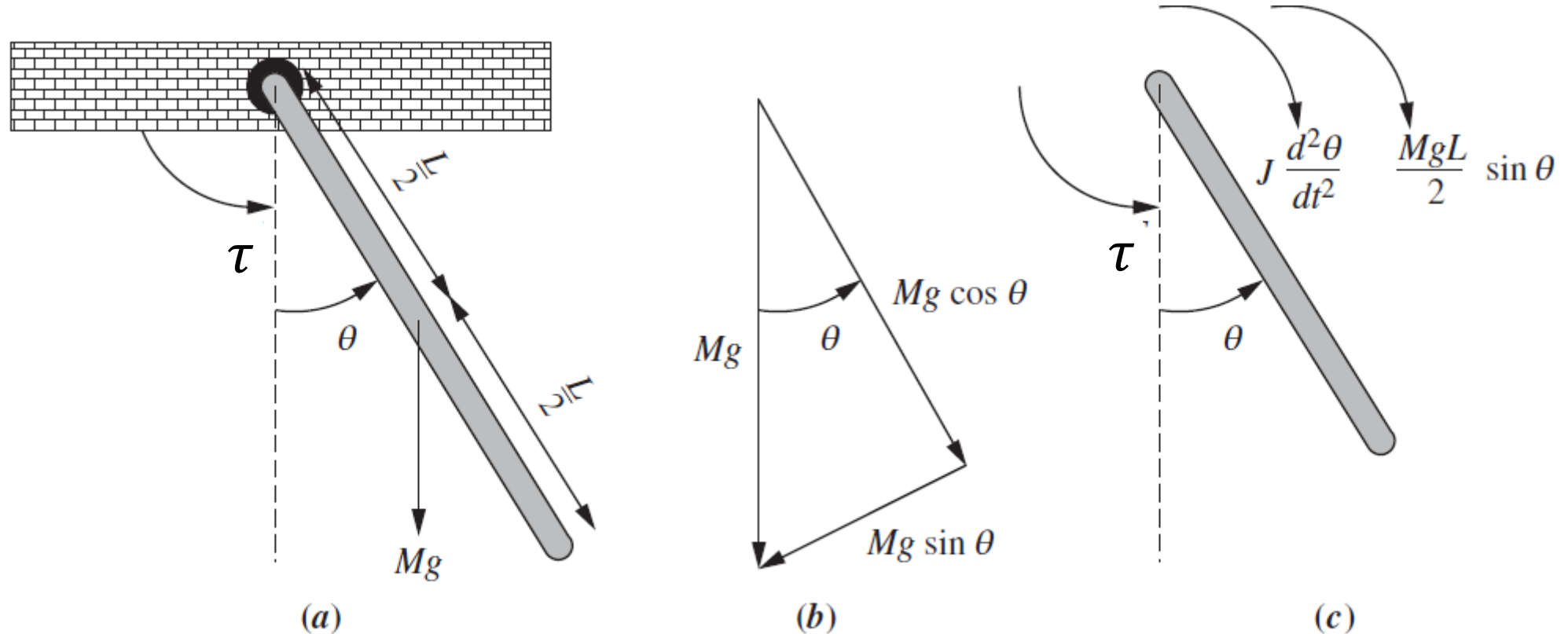


FIGURE 3.14 a. Simple pendulum; b. force components of Mg ; c. free-body diagram

Linearization

Ex. Transfer Function—Nonlinear System, simple pendulum (2)

Soln. According to the free-body diagram, the motion equation is:

$$J \frac{d^2 \theta}{dt^2} + \frac{MgL}{2} \sin \theta = \tau(t)$$

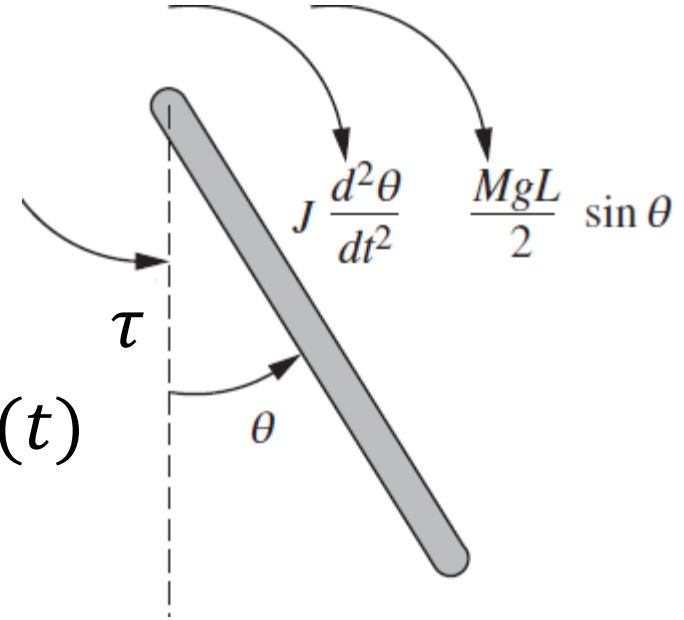
$$\theta = \theta_0 + \delta\theta \Rightarrow J \frac{d^2(\theta_0 + \delta\theta)}{dt^2} + \frac{MgL}{2} \sin(\theta_0 + \delta\theta) = \tau(t)$$

Linearize $\sin(\theta_0 + \delta\theta)$ when $\theta_0 = 0$.

$$\sin(\theta_0 + \delta\theta) - \sin(\theta_0) = \left. \frac{d \sin(\theta)}{d\theta} \right|_{\theta = \theta_0} (\theta - \theta_0) = \cos(\theta_0) \delta\theta = \delta\theta$$

Thus,

$$J \frac{d^2(\delta\theta)}{dt^2} + \frac{MgL}{2} \delta\theta = \tau(t) \xrightarrow{\mathcal{L}} Js^2 \delta\theta + \frac{MgL}{2} \delta\theta = T(s)$$



Linearization

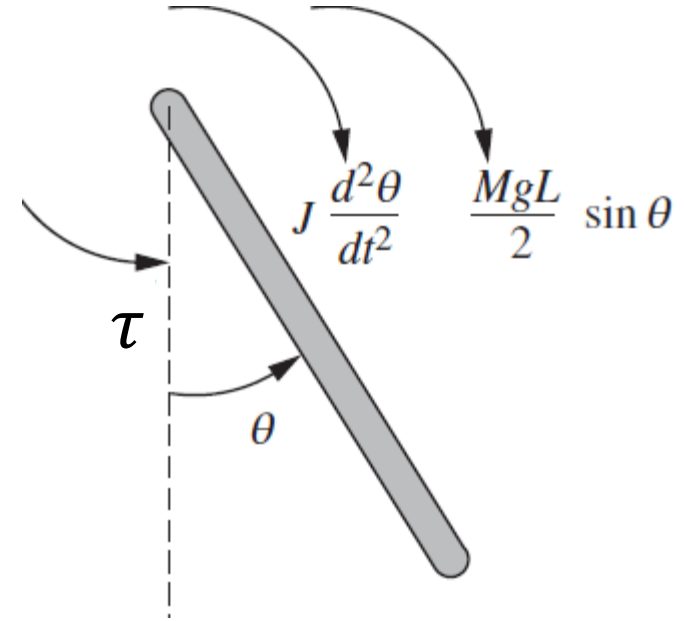
Ex. Transfer Function—Nonlinear System, simple pendulum (3)

Soln. (Cont.). Having

$$Js^2\delta\theta + \frac{MgL}{2}\delta\theta = T(s)$$

$$\Rightarrow \frac{\delta\theta(s)}{T(s)} = \frac{\frac{1}{J}}{s^2 + \frac{MgL}{2J}}$$

$$\Rightarrow \frac{\theta(s)}{T(s)} = \frac{1/J}{s^2 + \frac{MgL}{2J}}$$



for small excursions about $\theta_0 = 0$, or equivalently $\tau(t) = 0$.

Linearization in State Space

Ex. State Space—Nonlinear System (simple pendulum)

Soln. The motion equation is:

$$J \frac{d^2 \theta}{dt^2} + \frac{MgL}{2} \sin \theta = T$$

Letting $x_1 = \theta$, and $x_2 = \frac{d\theta}{dt}$ the state equations are

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -\frac{MgL}{2J} \sin x_1 + \frac{T}{J} \end{cases}$$

Given equilibrium points

$$x_1 = 0, \text{ and } x_2 = 0$$

$$x_1 = 0 + \delta x_1$$

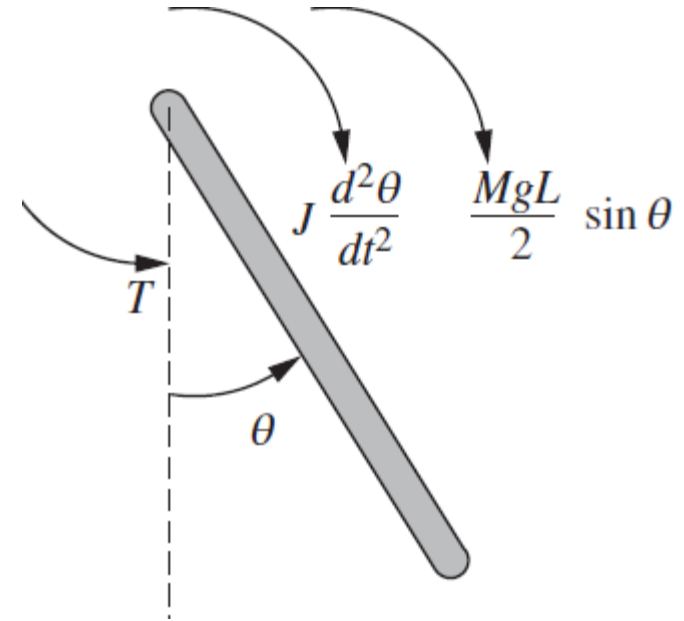
$$x_2 = 0 + \delta x_2$$

We have:

$$\sin x_1 - \sin 0 = \left. \frac{d(\sin x_1)}{dx_1} \right|_{x_1=0} \delta x_1 = \delta x_1 \Rightarrow \sin x_1 = \delta x_1$$

Thus, linearized state equations are

$$\begin{cases} \dot{\delta x}_1 = \delta x_2 \\ \dot{\delta x}_2 = -\frac{MgL}{2J} \delta x_1 + \frac{T}{J} \end{cases}$$



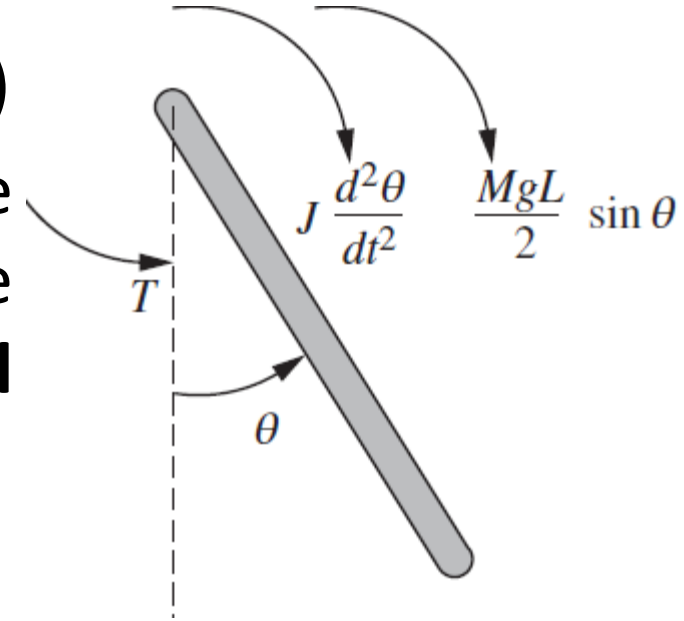
Linearization in State Space

Ex. State Space—Nonlinear System, simple pendulum (2)

Note. It is interesting to note that the nonlinear state equations represent a valid and **complete** model of the pendulum in state space even under **nonzero initial conditions** and even if parameters are **time varying**.

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -\frac{MgL}{2J} \sin x_1 + \frac{T}{J} \end{cases}$$

However, if we want to apply classical techniques and convert these state equations to a transfer function, we must linearize them.



Linearization in State Space

It is interesting to note

that the nonlinear Eqs. (3.80) represent a valid and complete model of the pendulum in

state space even under nonzero initial conditions and even if parameters are time varying.

However, if we want to apply classical techniques and convert these state equations to a

transfer function, we must linearize them.